

Zirui (Ken) Deng

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Summary: PhD candidate in Computer Science with strong experience applying machine learning and statistical methods to large-scale datasets. Skilled in designing experiments, analyzing data, and building models to drive product and business decisions. Experience working with real-world data to extract insights, evaluate performance, and improve system outcomes. Experience with sequence modeling and prediction tasks relevant to recommendation systems and user modeling. Interested in applying data science and machine learning to large-scale user-facing systems.

Skills:

Machine Learning — Deep learning, predictive modeling, anomaly detection, time-series modeling

Statistics & Experimentation — Hypothesis testing, regression, experimental design, A/B testing

Data Science — Exploratory data analysis, feature engineering, model evaluation, SQL-based analysis

Data Engineering — ETL pipelines, large-scale data processing, data validation

Programming — Python (NumPy, Pandas, scikit-learn, PyTorch, TensorFlow), SQL

Tools — Git, Linux

EDUCATION

Washington University in St. Louis

PhD Candidate in Computer Science

St. Louis, Missouri

August 2021 – May 2027 (anticipated)

- Focus: Information theory, statistical inference, machine learning, quantitative modeling
- Relevant Areas: Probability, optimization, statistical inference, machine Learning, algorithms

Emory University

B.S. in Applied Mathematics and Statistics & B.S. in Computer Science

Highest Honors in Applied Mathematics and Statistics

Atlanta, Georgia

August 2017 – May 2021

·Major GPA: 3.92 / 4.0

·GPA: 3.8 / 4.0

·Relevant Coursework: Numerical Analysis, Multi-variable Calculus, Linear Algebra, Optimization, Partial Differential Equations, Probability & Statistics, Data Structures and Algorithms, Database Systems, Systems Programming, Deep Learning, Machine Learning

·2-time Putnam Mathematical Competition participant

RESEARCH EXPERIENCE

Predicting Surgical Case Duration Using Neural ODEs

Washington University in St. Louis

St. Louis, Missouri

August 2021 – November 2021

- Built time-series prediction models (LSTM, Neural ODE) on large-scale real-world datasets.
- Performed exploratory data analysis to identify key patterns and drivers of outcomes.
- Designed and conducted experiments to compare model architectures and evaluate performance trade-offs.
- Reduced forecasting error by 15% (CRPS), improving system performance and decision-making.
- Analyzed model behavior to generate insights for real-world applications.

Undergraduate Honors thesis: Fair algorithms in machine learning

Atlanta, Georgia

Emory University

May 2020 – May 2021

- Conducted theoretical and empirical analysis of fairness constraints using statistical and optimization frameworks.
- Studied trade-offs between competing objectives under constraints, with applications to robust decision systems.
- Evaluated clustering and learning algorithms under varying assumptions to test stability and robustness.

Deep Learning Project: Multi-variate Neural Ordinary Differential Equations for Time-series Modeling

Atlanta, Georgia

Remote from Emory University

May 2020 – July 2020

- Implemented multiple deep learning architectures (CNNs, GRUs, LSTMs) for time-series modeling.
- Designed reusable training components and optimized model performance through experimentation.

Machine Learning Project on Credit Card Fraud

Atlanta, Georgia

Emory University

Nov 2019 – Dec 2019

- Analyzed transactional datasets to identify behavioral patterns and anomalies in transactional data, analogous to user activity modeling.
- Built predictive models for rare-event detection and evaluated performance using precision-recall metrics.
- Identified key features driving model performance and predictive signals relevant to ranking and classification tasks.
- Generated insights to support decision-making in real-world systems.

INTERNSHIPS

Data Analyst Internship

China Telecom

Nanjing, China

May 2019 – June 2019

- Built and maintained data pipelines for automated large-scale data collection.
- Implemented data cleaning and validation processes to ensure high-quality datasets.
- Improved efficiency and reliability of data workflows used in analytics systems.
- Ensured data quality and robustness through validation and preprocessing pipelines.
- Delivered structured analyses to support data-driven decision-making.

Investment Banking Analysis Internship

Ten Billion Research Laboratory in Shanghai

Shanghai, China

May 2018 – June 2018

- Performed quantitative research and SWOT analyses on blockchain technologies; translated qualitative findings into detailed, data-driven reports.
- Produced structured analyses and reports to communicate findings and insights.
- Engaged in regular trend analysis meetings, using quantitative forecasting to predict industry shifts and opportunities.

PUBLICATIONS (Information Theory & Statistical Inference — uncertainty quantification in ML systems)

Z. Deng, V. Ramkumar, N. Raviv, “Private Inference in Quantized Models,” IEEE Transactions on Information Theory, vol. 71, no. 5, pp. 3957- 3973, 2025.

Z. Deng, V. Ramkumar, N. Raviv, “Perfect Subset Privacy in Polynomial Computation,” IEEE International Symposium on Information Theory (ISIT), 2024.

Z. Deng, N. Raviv, “Approximate Private Inference in Quantized Models,” IEEE International Symposium on Information Theory (ISIT), 2023.

ADDITIONAL INFORMATION

Languages: Fluent in English and Mandarin Chinese

Interests: Poker; Weiqi (national second-level athlete)